

Overview of Clouds



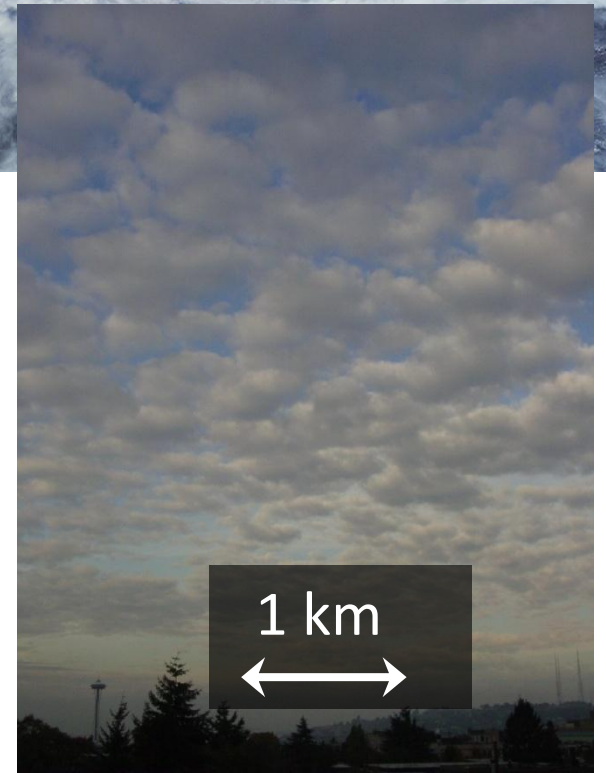
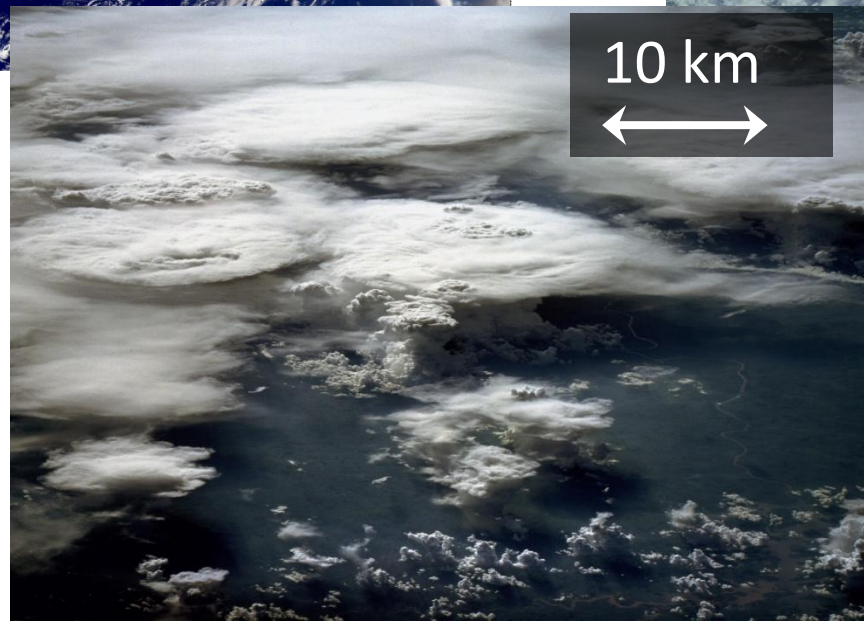
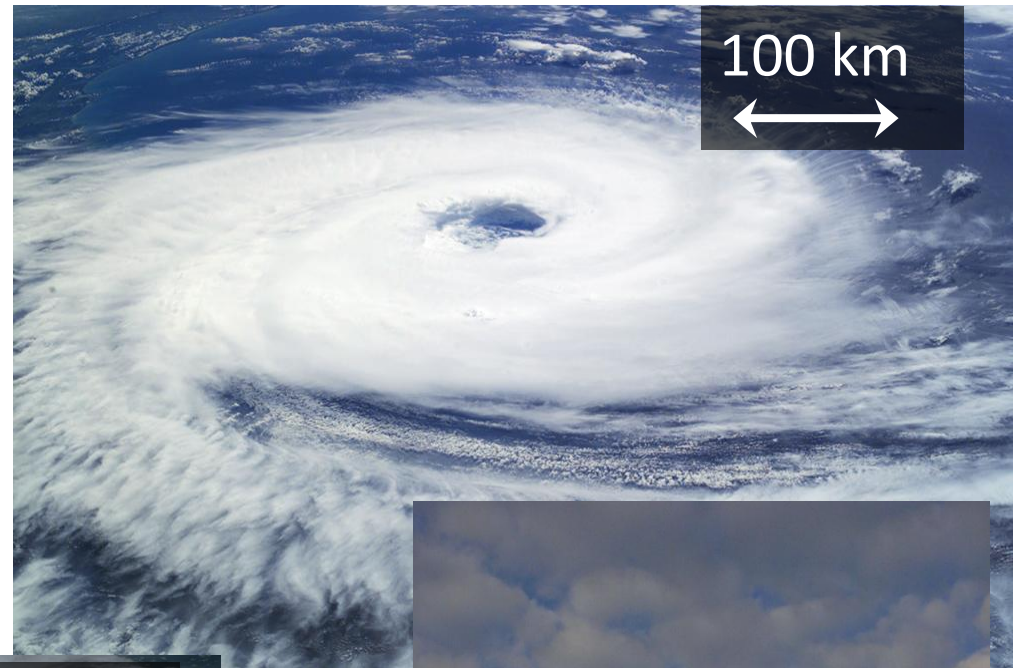
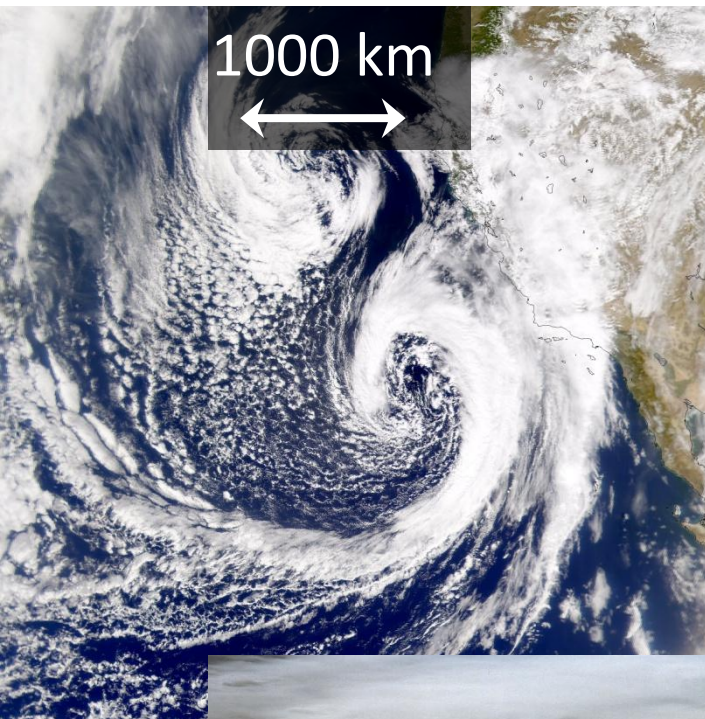
Cloud

- ☞ A cloud is aggregate of tiny water droplets or ice crystals suspended in the air.
- ☞ Some are found at high elevations, whereas others nearly touch the ground.

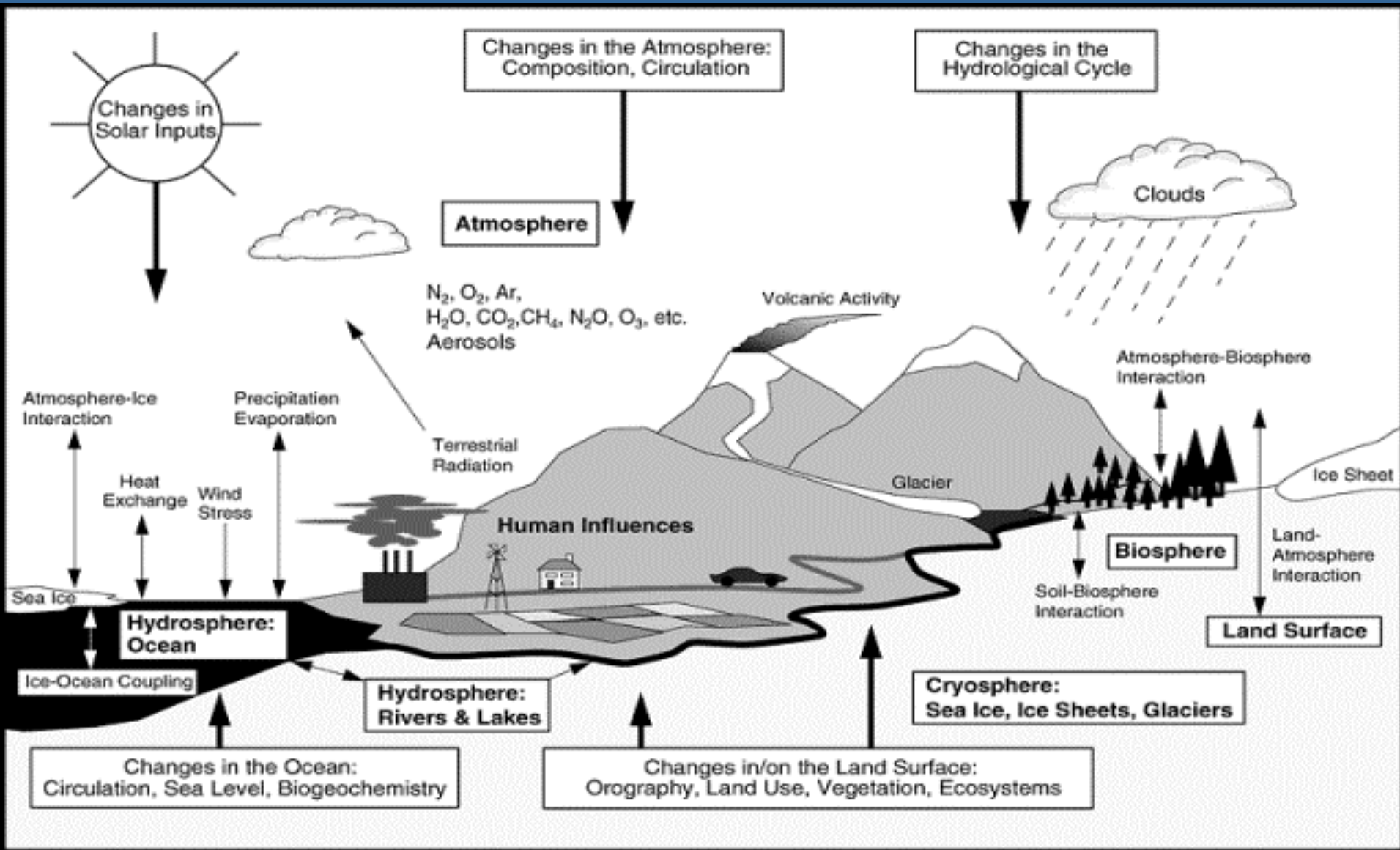
Why are we interested in clouds?

- Clouds are a major factor in the **earth's radiation budget** because they interact strongly with both short and long-wave radiation.
- Clouds are a key step in the **hydrological cycle**.
- Clouds affect significantly **vertical transport and redistribution** of species in the atmosphere.

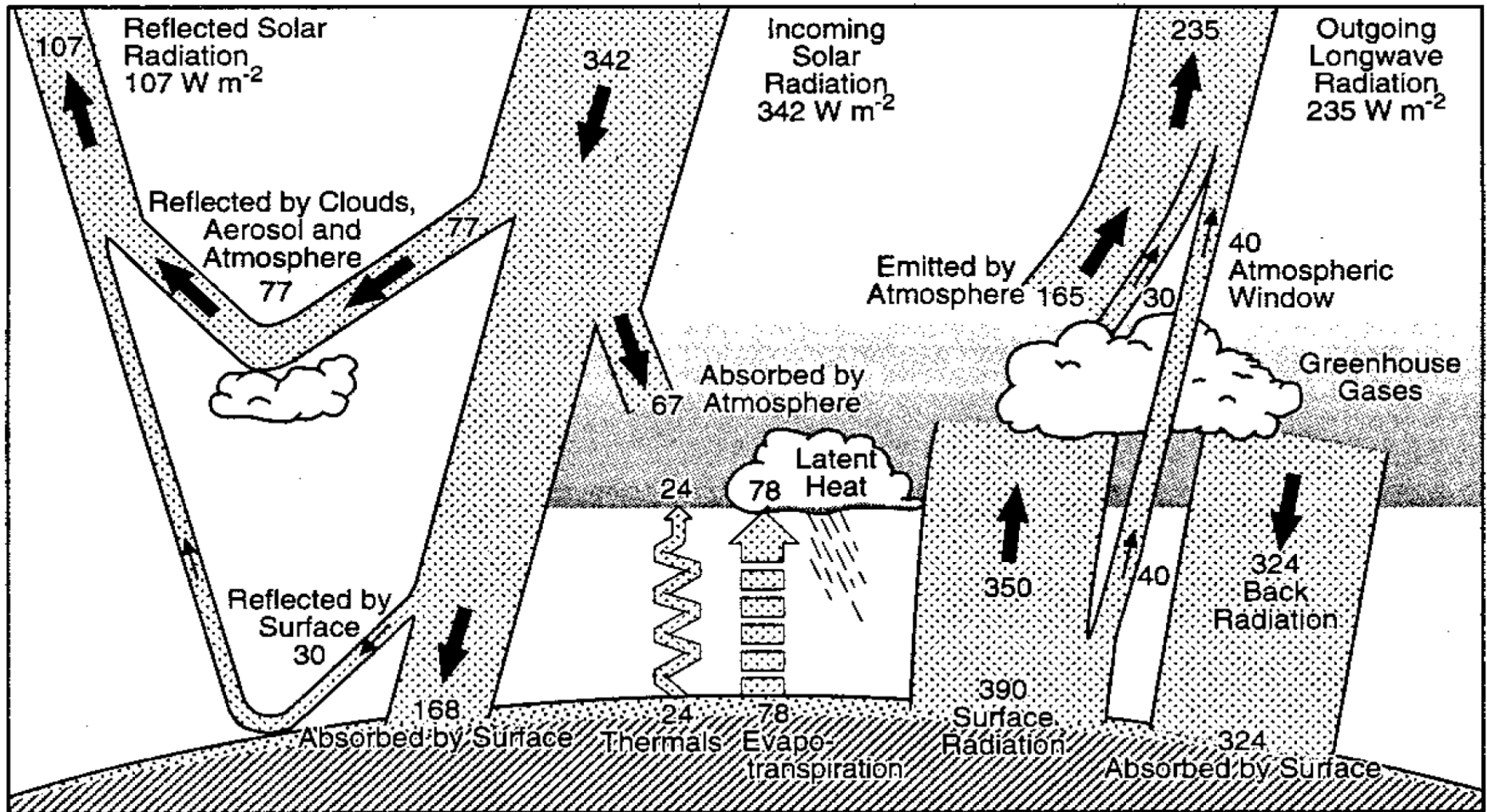
From global to local – range of scales



The Global Climate System



Radiation in the climate system



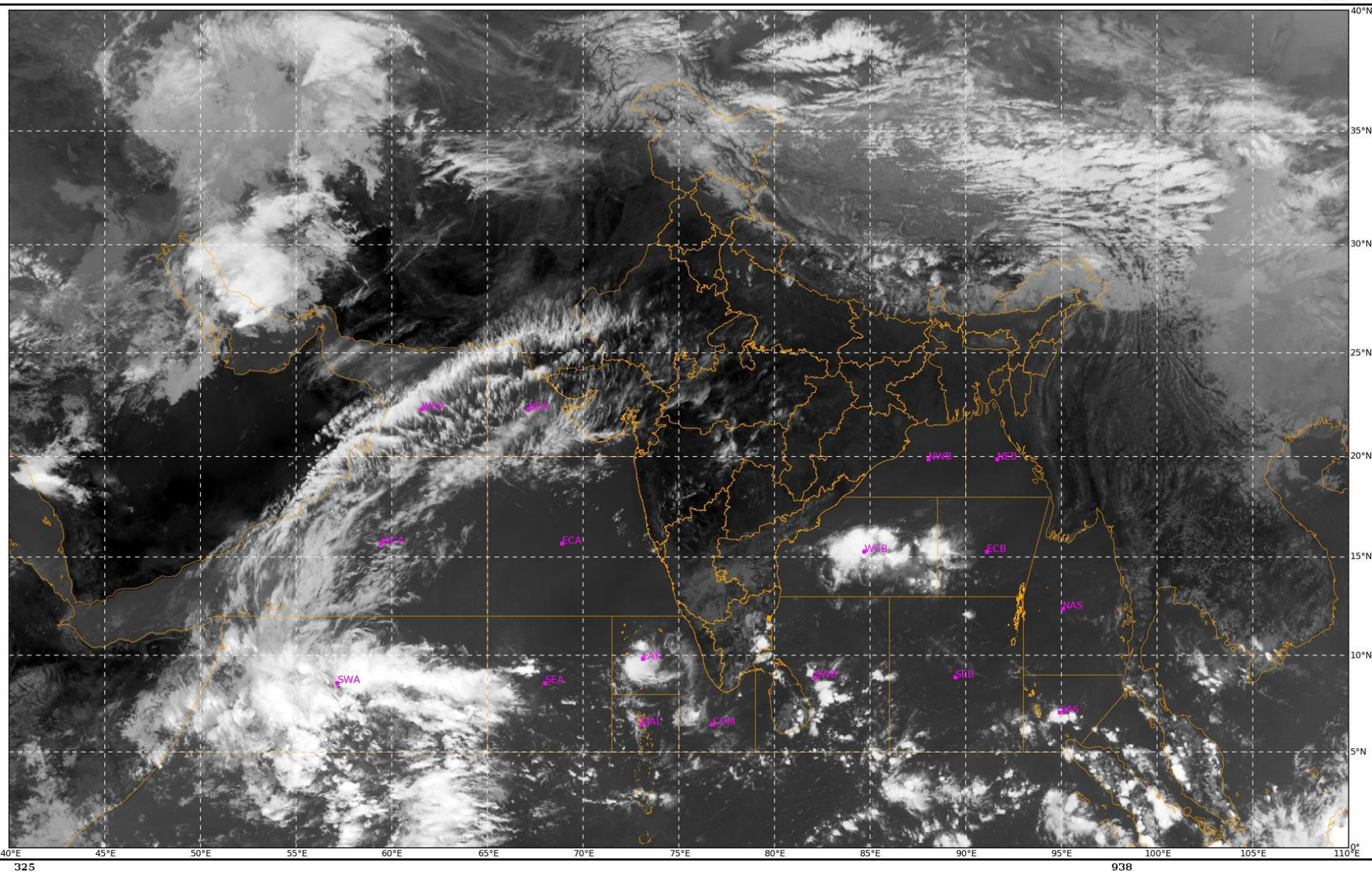
SAT : INSAT-3DR IMG

IMG_TIR1 10.8 um

L1C Mercator

18-11-2019/(0744 to 0811) GMT

18-11-2019/(1314 to 1341) IST



IMD, DELHI

The IR pattern emphasizes the higher cloud tops, which largely determine the precipitation pattern.



Atmospheric Stability

☞ **Atmospheric stability** – a measure of the atmosphere's susceptibility to vertical motion.

Atmospheric stability
comes in 3 flavors:

➤ **Stable**

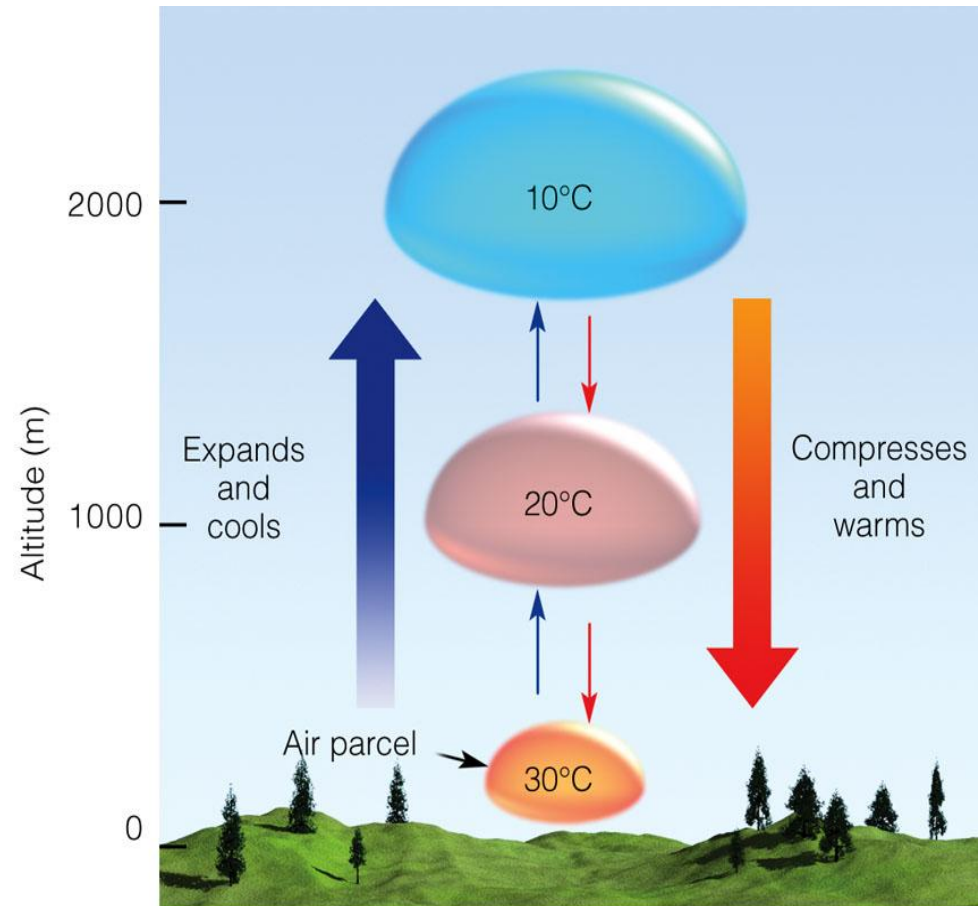
➤ **Unstable**



➤ **Neutral**

Basically

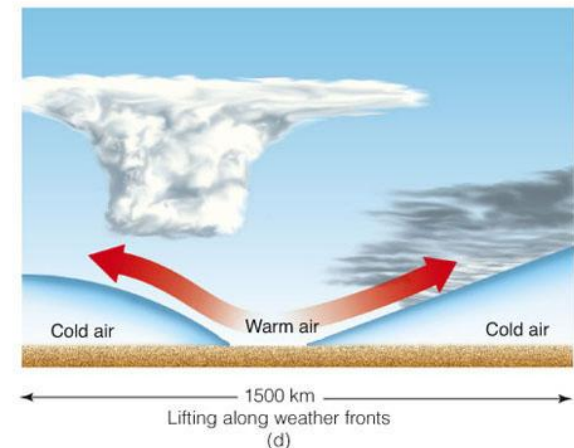
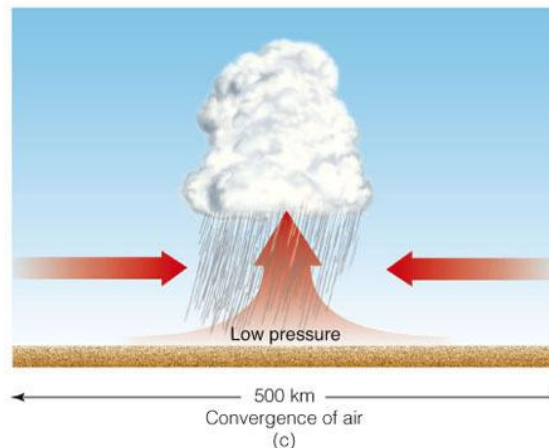
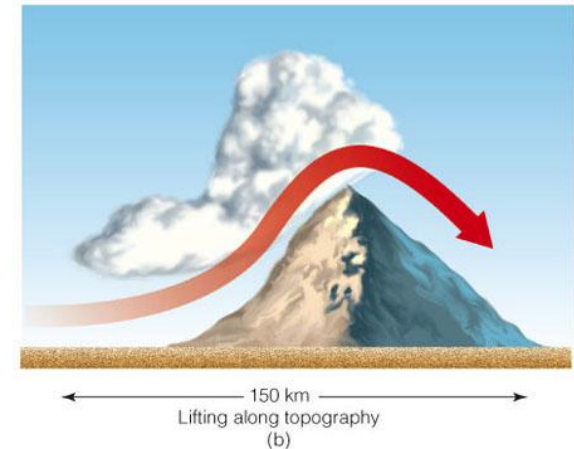
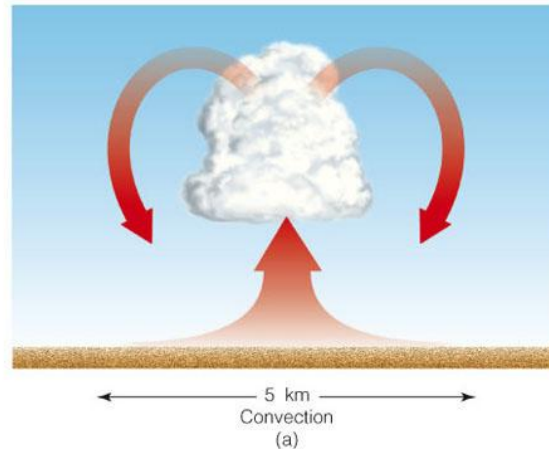
- ❖ Lifting of air leads to **cooling**
- ❖ **Cooling** leads to higher relative humidity
- ❖ Once saturation is reached, droplets/**CLOUD** can form



Four Ways for Cloud Development

Mechanisms

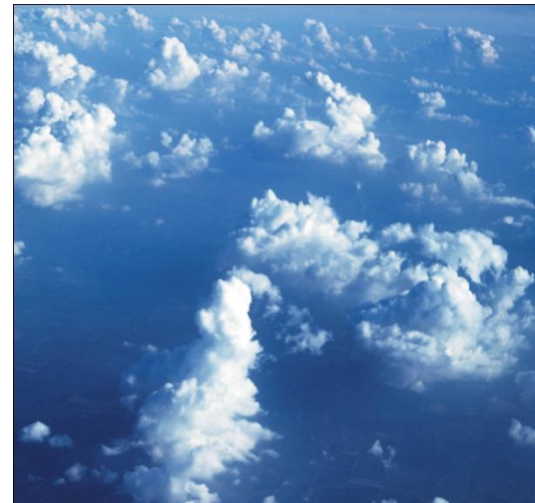
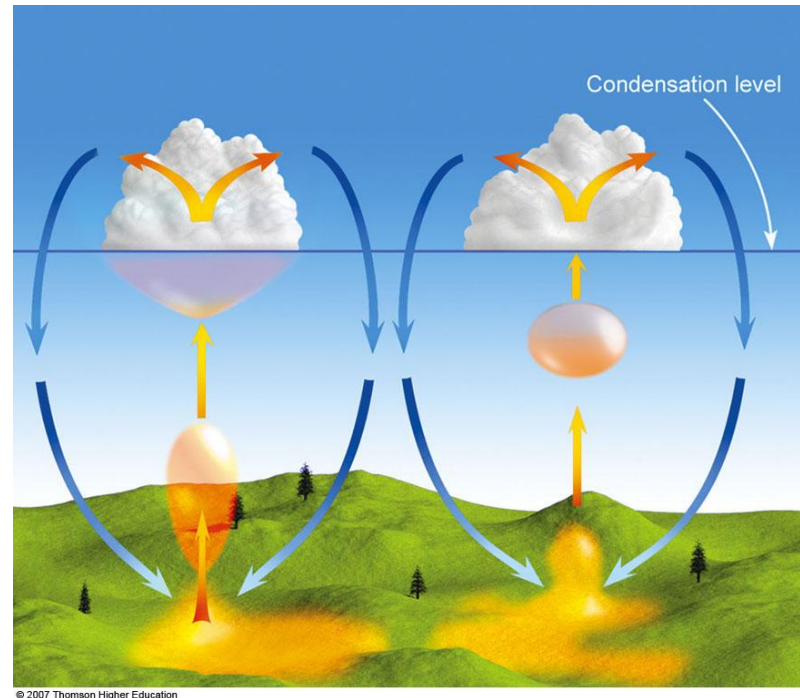
- Surface Heating and Free Convection
- Uplift along Topography
- Convergence of Air
- Lifting Along Weather Fronts



Surface Heating-Convection

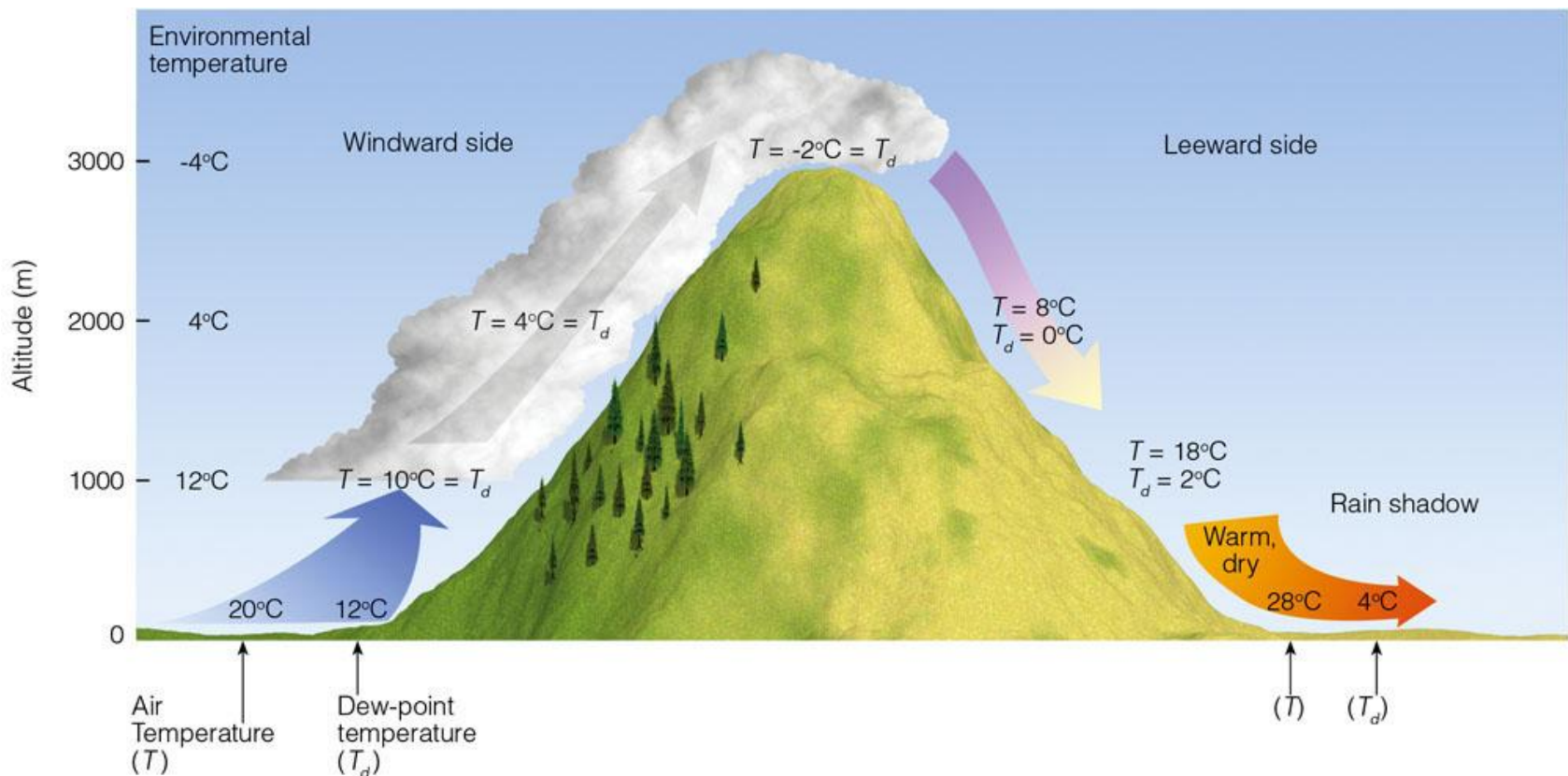
- Thermals
- Fair weather cumulus

- ❖ Fair weather cumulus provide a visual marker of thermals.
- ❖ Bases of fair-weather cumulus clouds marks the lifting condensation level, the level at which rising air first becomes saturated.

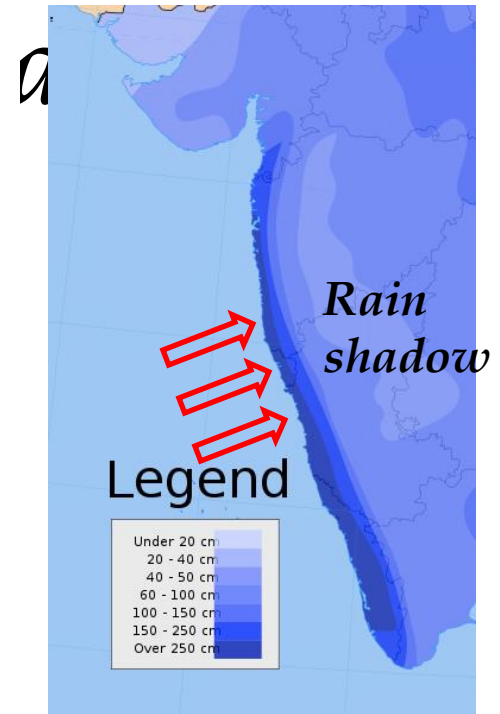
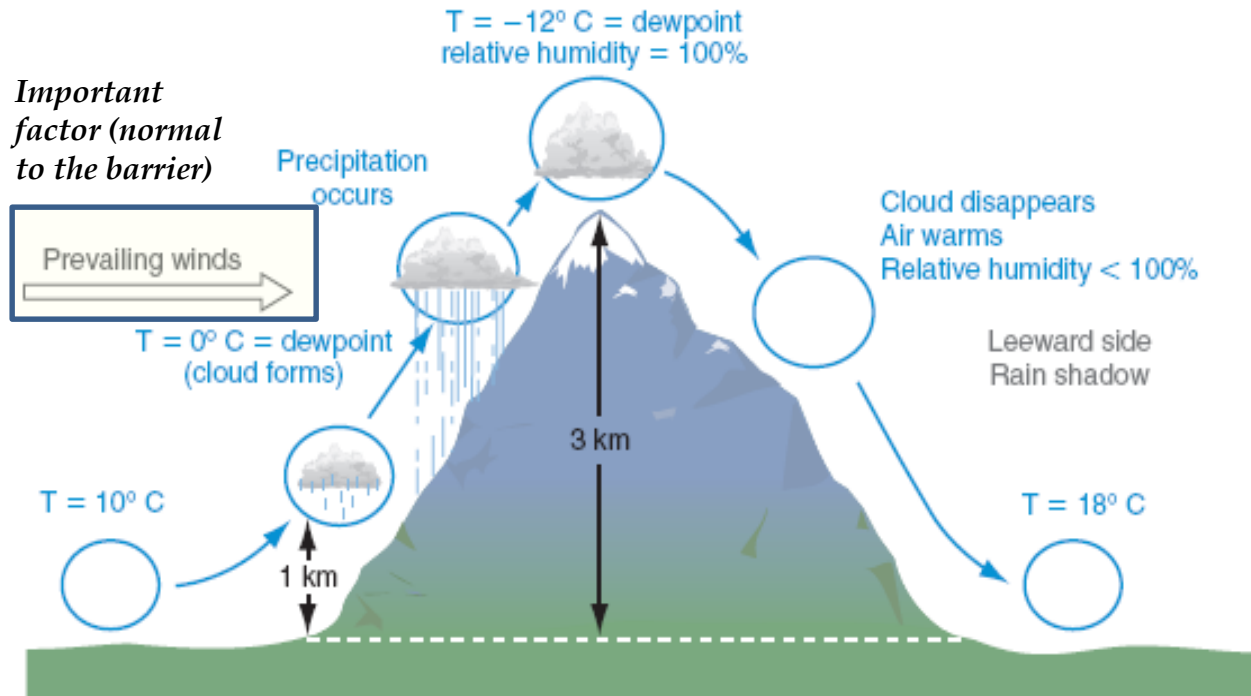


Topography induced Clouds

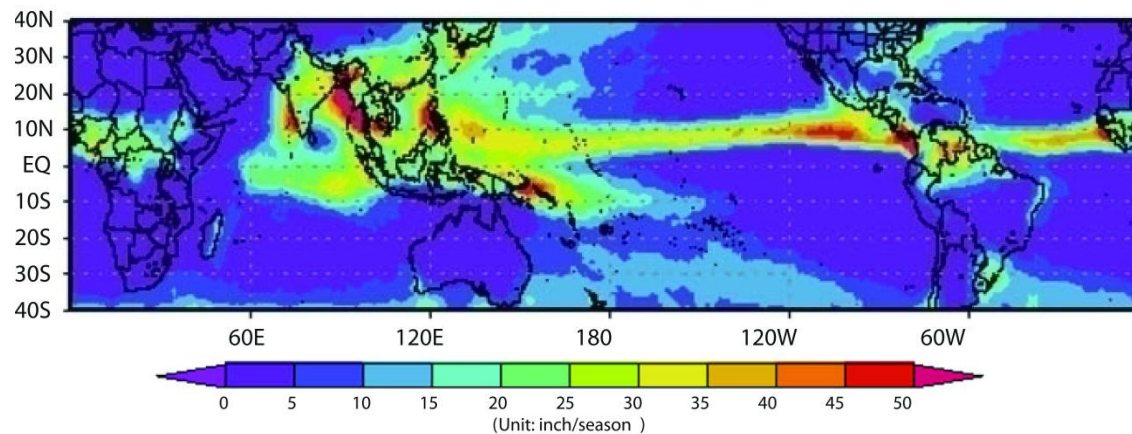
- Rain shadow: Due to frequent westerly winds, the western slope of the Rocky Mountains receives much more precipitation than the eastern slope.



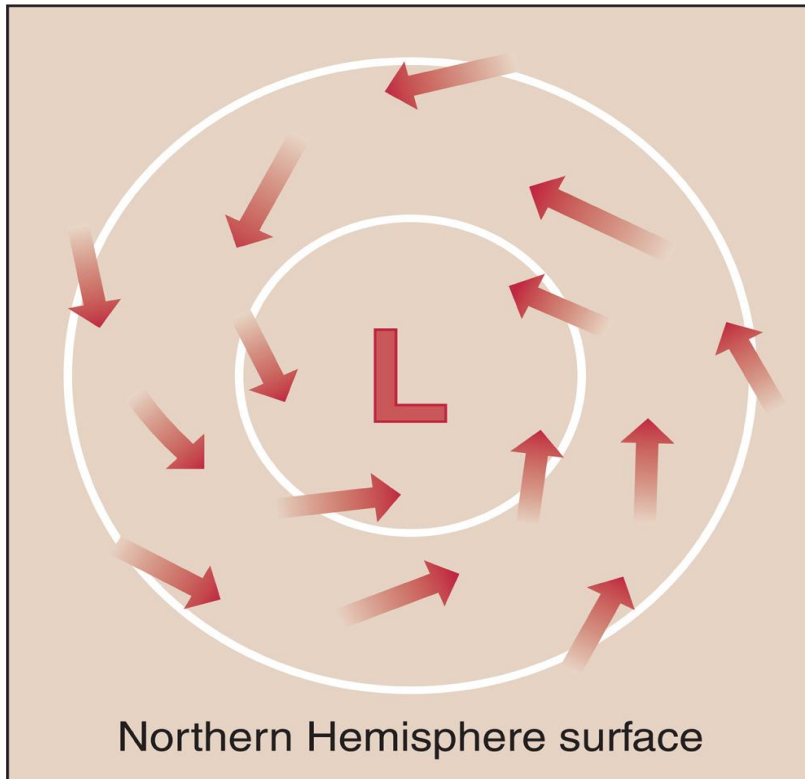
Topography and Clouds



3B43 Seasonal Rainfall Climatology (JJA 1998 - 2003)



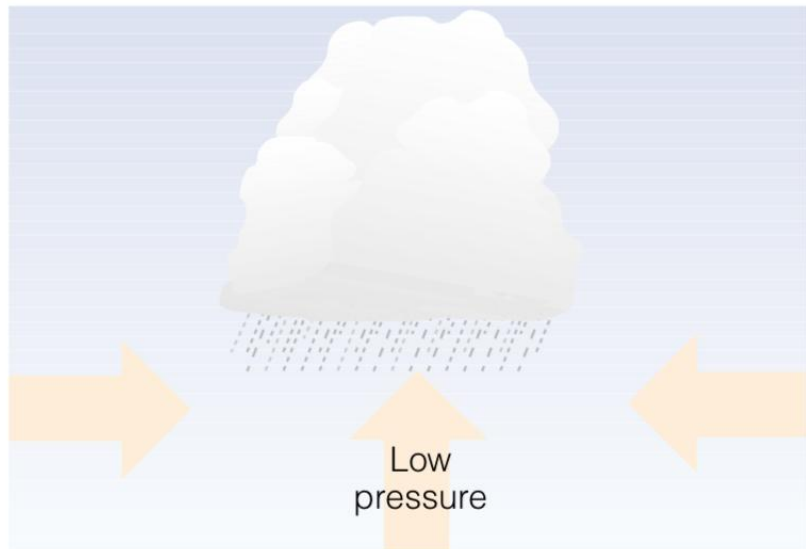
Convergence



(a)

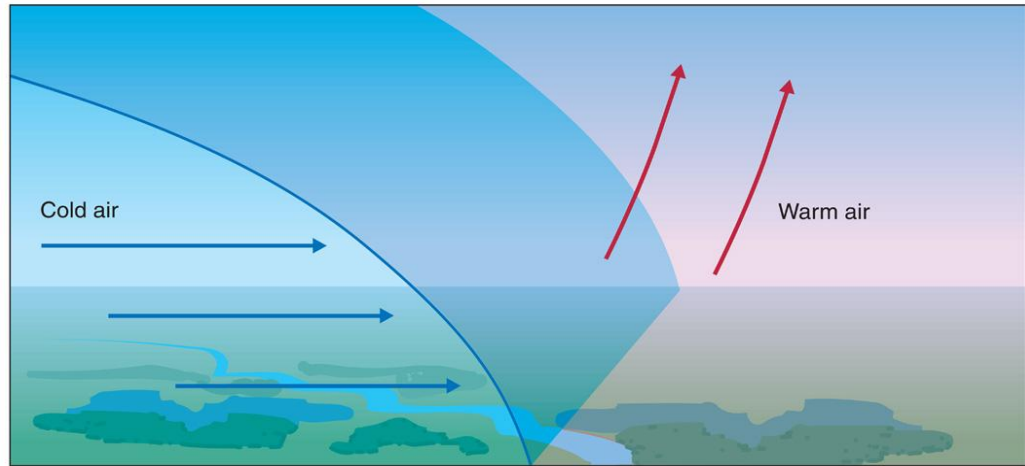
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“Air must rise when it converges”



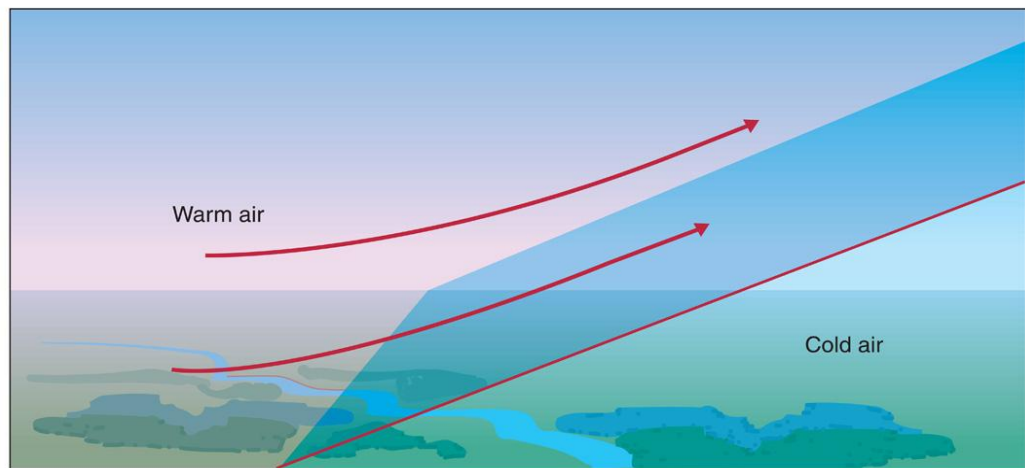
Frontal Lifting

Cold Front
(cold air pushes warm air up)



(a)

Warm Front
(Warm air overruns cold air)



(b)

Cloud Classification

Based on type:

- 1) Cirrus (high, thin, wispy)
- 2) Stratus (layered)
- 3) Nimbus (rain-producing)
- 4) Cumulus (puffy, vertically-developed)

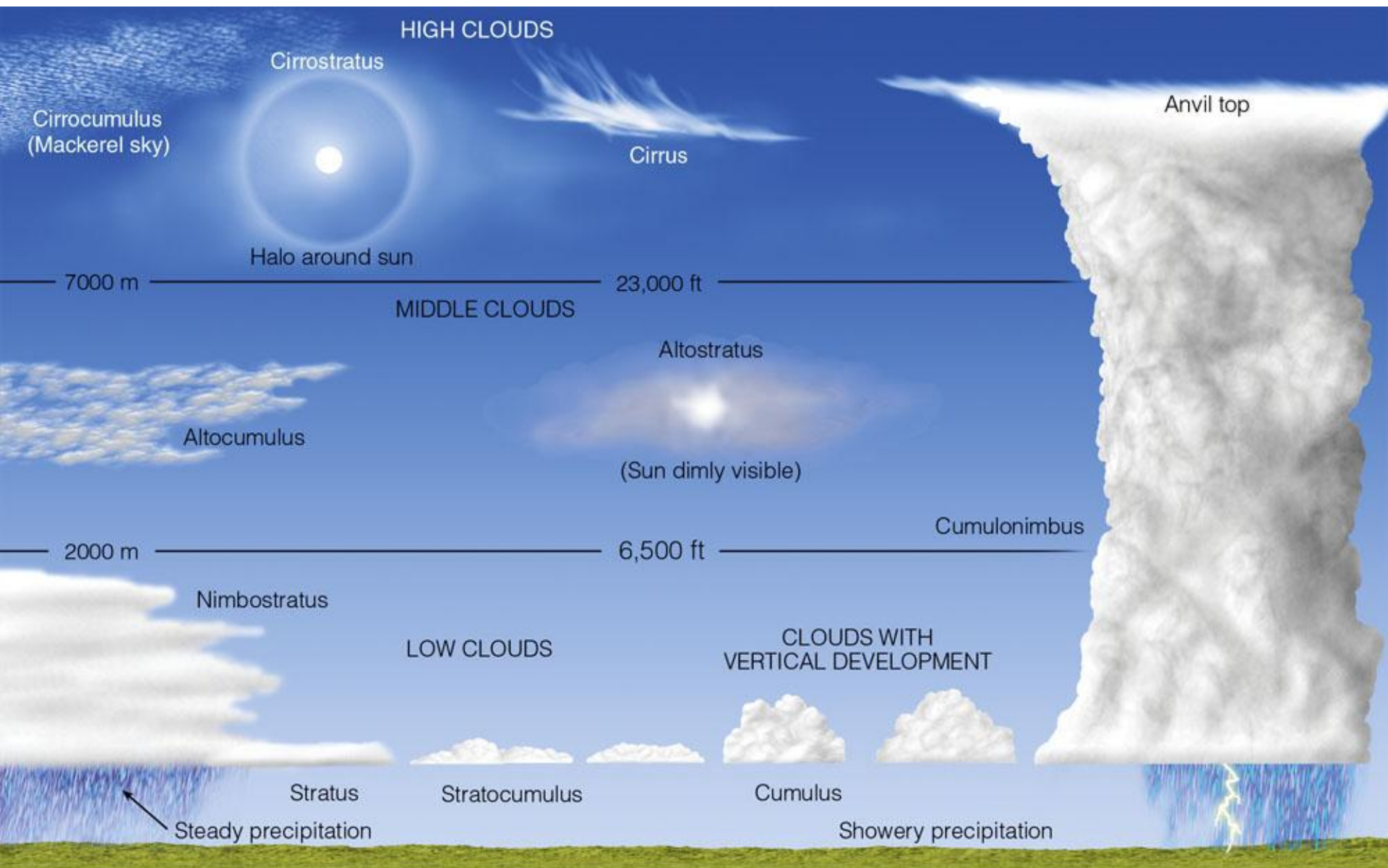
Based on Hight:

- High clouds (higher than 19,000 ft.)
- Middle clouds (b/w 6,000 and 19,000 ft.)
- Low clouds (below 6,000 ft.)
- Clouds with vertical development

1. High clouds	3. Low clouds
Cirrus (Ci)	Stratus (St)
Cirrostratus (Cs)	Stratocumulus (Sc)
Cirrocumulus (Cc)	Nimbostratus (Ns)
2. Middle clouds	4. Clouds with vertical development
Altostratus (As)	Cumulus (Cu)
Alto cumulus (Ac)	Cumulonimbus (Cb)

Cloud Group	Tropical Region	Middle Latitude Region	Polar Region
High Ci, Cs, Cc	20,000 to 60,000 ft (6000 to 18,000 m)	16,000 to 43,000 ft (5000 to 13,000 m)	10,000 to 26,000 ft (3000 to 8000 m)
Middle As, Ac	6500 to 26,000 ft (2000 to 8000 m)	6500 to 23,000 ft (2000 to 7000 m)	6500 to 13,000 ft (2000 to 4000 m)
Low St, Sc, Ns	surface to 6500 ft (0 to 2000 m)	surface to 6500 ft (0 to 2000 m)	surface to 6500 ft (0 to 2000 m)

CLOUD CLASSIFICATION



Cumulus congestus



Stratocumulus



Cumulonimbus



Billow clouds



Fair weather cumulus

Cirrus



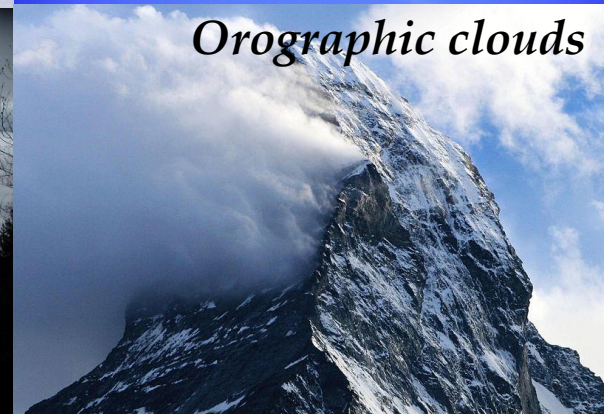
Alto cumulus



Stratus/fog



Orographic clouds



High Level Clouds

- ☞ The most common form of high-level clouds are thin and often cirrus clouds.
- ☞ High-level clouds form above 20,000 feet (6,000 meters) and since the temperatures are so cold at such high elevations, these clouds are primarily composed of ice crystals.
- ☞ High-level clouds are typically thin and white in appearance.

Cirrus generally occur in fair weather.

e.g. Cirrus Clouds





Sometimes the only indication of their presence is given by an observed halo around the sun or moon. Halos result from the refraction of light by the cloud's ice crystals

Cirrostratus

Cirrocumulus clouds appear as small, rounded white puffs.

The small ripples in the cirrocumulus sometimes resemble the scales of a fish.

A sky with cirrocumulus clouds is sometimes referred to as a "mackerel sky."



Cirrocumulus

More examples of Cirrocumulus



Middle Level Clouds

- ☛ The bases of mid-level clouds typically appear between 6,500 to 20,000 feet (2,000 to 6,000 meters).
- ☛ Because of their lower altitudes, they are composed primarily of water droplets, however, they can also be composed of ice crystals when temperatures are cold enough.

e.g. Altocumulus Clouds



Altostratus



- They are more extensive and composed primarily **water droplets & Ice crystals**.
- Scatter a large proportion of **incoming sunlight** back to space.
- These clouds usually cover the entire sky. The isolation that does make its way to the surface consists primarily or exclusively as diffuse radiation.
- When viewing the Sun or Moon behind altostratus, one sees a bright spot behind the clouds instead of a halo.



Altostratus

Altostratus are potentially dangerous to aircraft because they can cause ice accumulation on the wings

Alto cumulus



- ➡ They are layered clouds that form long bands or contain a series of puffy clouds arranged in rows and consist mainly of liquid droplets rather than ice crystals.
- ➡ located between 6,500 and 20,000 feet (2,000 to 6,000 meters) above the ground.



Low Level Clouds

- ☞ Low clouds are mostly composed of water droplets since their bases generally lie below 6,500 feet (2,000 meters).
 - ☞ However, when temperatures are cold enough, these clouds may also contain ice particles and snow.
 - ☞ Stratus clouds are horizontal, layered clouds that stretch out across the sky like a blanket.
- e.g. Nimbostratus Clouds dark, low-level clouds with precipitation,

Stratus

They resemble fog that does not reach the ground. Usually no precipitation falls from stratus clouds, but sometimes they may drizzle.



Nimbostratus



- ❖ Low, layered clouds that yield **moderate to heavy precipitation** are called nimbostratus. Precipitation generally **steady and prolonged**. Referred as “Rainy day clouds”.
- ❖ Seen from below, these clouds look very much like stratus, except for the presence of precipitation.
- ❖ It is **a thick, wet blanket** with a ragged base caused by the continual precipitation.
- ❖ These are composed of **Liquid water, raindrops, snowflakes and ice crystals**

Stratocumulus



- Low, layered clouds with **some vertical development**. Their darkness varies when you see from below - because their thickness varies across the cloud.
- **Thicker sections appear dark**, and **thinner areas appear as bright** spots. They join into a semi-continuous layer.
- Mainly composed of **liquid water** and usually formed from cumulus or stratus clouds.

Vertically Developed clouds

- ☛ Probably the most familiar of the classified clouds is the cumulus cloud.
- ☛ Generated most commonly through either convection or frontal lifting, these clouds can grow to heights in excess of 12,000 meters, releasing incredible amounts of energy through the condensation of water vapor within the cloud itself.

e.g. Cumulonimbus



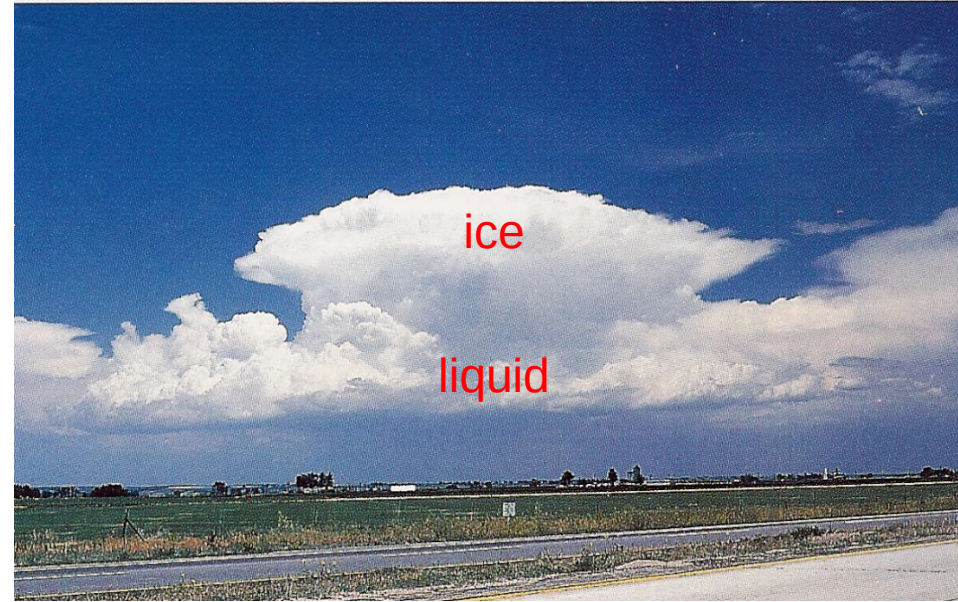


Cumulonimbus

- The most violent of all clouds and produce the most intense thunderstorms.
- In warm, humid, and unstable air, they can have bases just a few hundred meters above the surface and tops extending into the lower stratosphere.
- A cumulonimbus is distinguished by the presence of an anvil composed entirely of ice crystals formed by the high winds of the lower stratosphere that extend the cloud forward.
- They can exist as individual towers or form a line of towers called a squall line.

Cumulonimbus (Cb)

- ❑ Fueled by vigorous convective updrafts (sometimes in excess 50 knots), the tops of cumulonimbus clouds can easily reach 39,000 feet (12,000 meters) or higher.
- ❑ Can develop to great heights, often with anvil top.
- ❑ It produces heavy showers of rain-snow-hail, lightning-thunder, strong gusty surface winds.
- ❑ Liquid water throughout, ice crystals at the top.





Cumulus congestus

- Intensely developed clouds.
- They consist of multiple towers, and each tower has several cells of uplift.
- This gives them a fortress-like appearance with numerous columns of varying heights.
- Their strong vertical development implies that these clouds form in unstable air.

Other clouds types

Miscellaneous cloud types which do not fit into the previous four groups.



Mammatus (Pouch Like)



Contrails



Billow clouds



Orographic

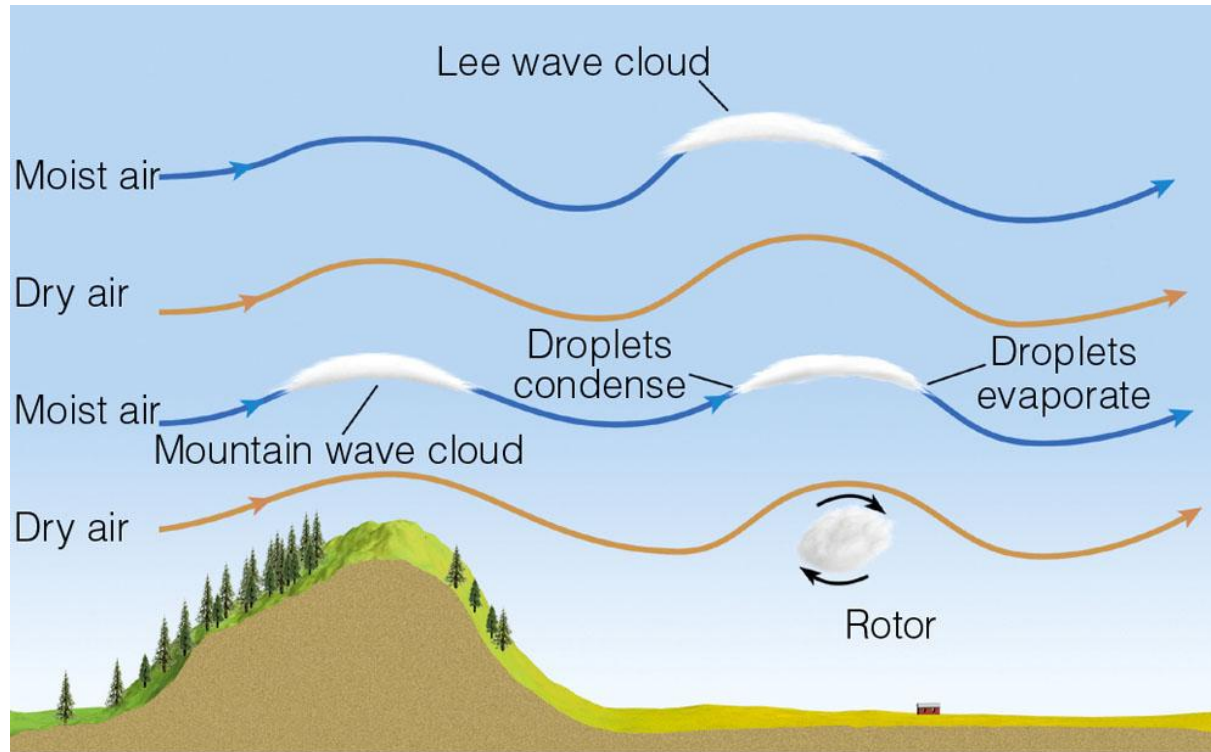


Lenticular (Lens)



Pileus (Skull Cap)

Wave clouds



*Cirriform clouds at 8.5 km over the Himalayas.
Lhotse Peak (8500 m, 29.68 N, 86.68 E)*



Lenticular (Lens)

Clouds life cycle

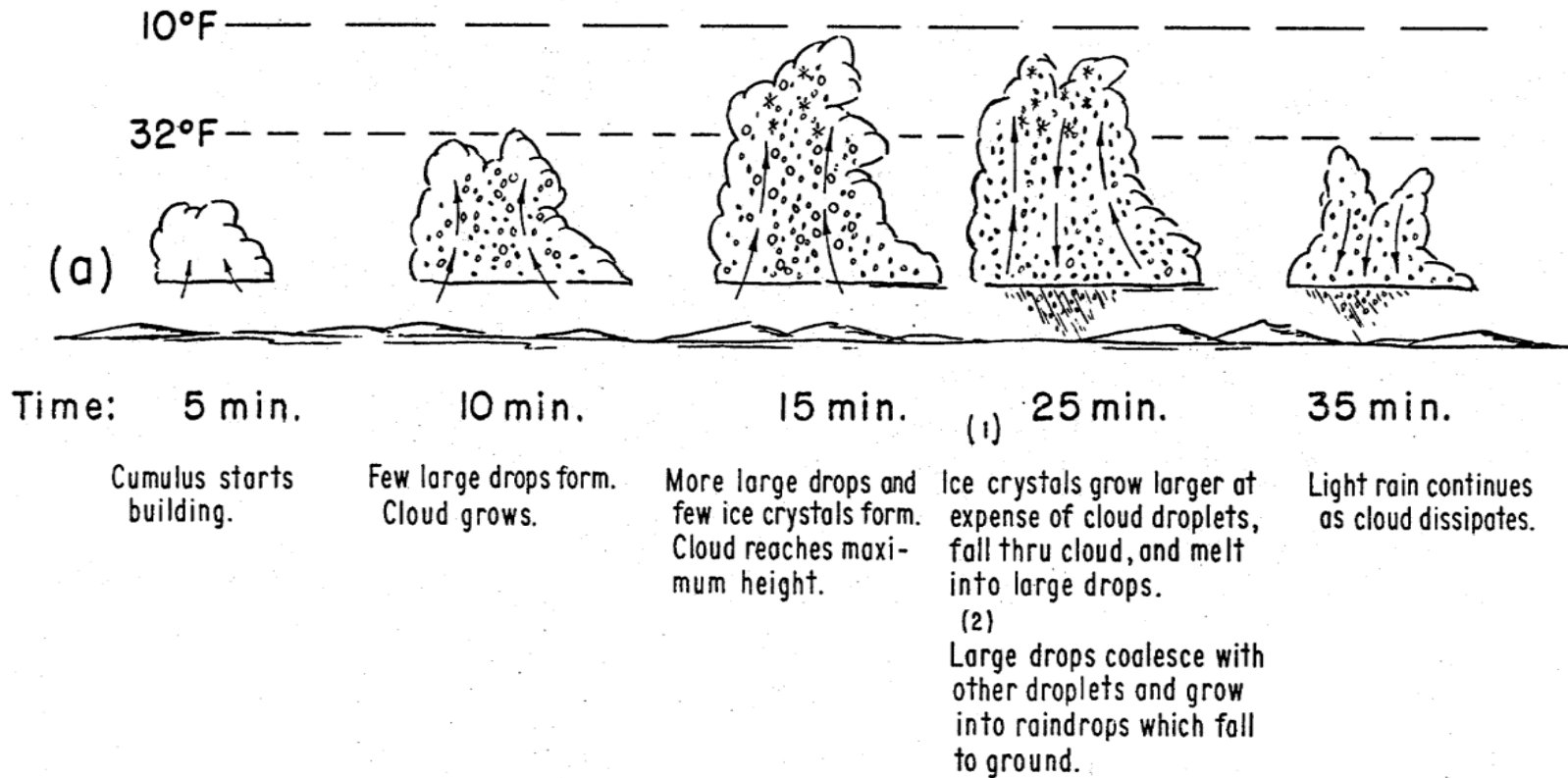
Just like human beings the clouds have a life cycle and they are born, they grow up, they get aged and ultimately die.

Small	Medium	Large
~5-10 mins	~ 30mins	> 1hour more

A large nimbus may process about 10 kilo-tonnes per second of water amounting to about 50,000 kilo-tonnes during its life span of about 1.5 hours.

LIFE CYCLE OF CLOUD DEVELOPMENT

NATURAL CLOUD



LEGEND

* -Ice

• -Liquid droplets

Precipitation Process

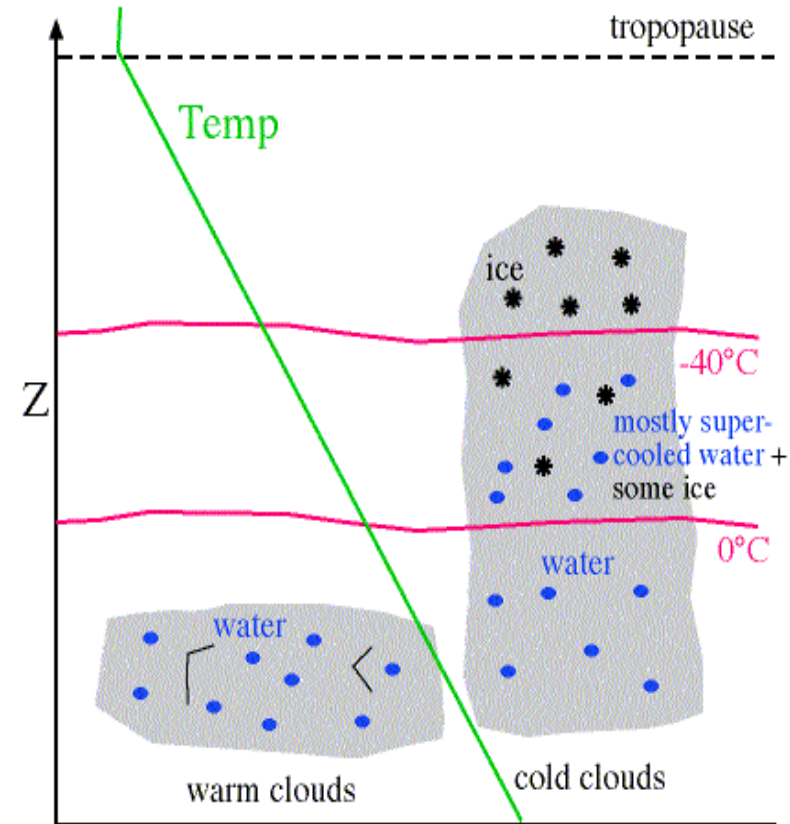
- When it comes to forming precipitation, there are two basic types of clouds

Warm Clouds: Clouds that lie completely below the 0°C isotherm, referred to as warm clouds.

- Containing only water droplets
- **Collision-Coalescence mechanism**

Cold Clouds: If a cloud extends above the 0°C level, it is called a cold cloud.

- Contains ice particles, supercooled droplets.
- Even though the temperature may be below 0°C , water droplets can still exist in clouds, in which case they are referred to as **supercooled droplets**.
- If contains both it is called mixed phase cloud. If entirely ice called glaciated.
- Dominant rain making process:
Bergeron-Findeisen process



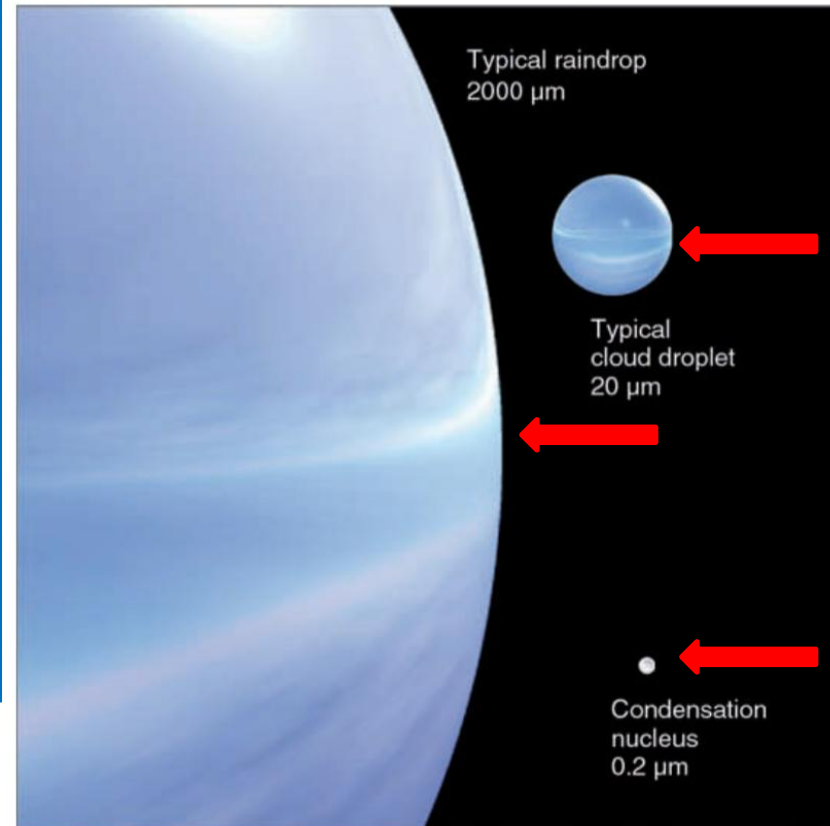
Relative sizes of raindrops, cloud droplets and condensation nuclei in micrometers (μm)

A typical rain drop is about 100 times larger than a typical cloud droplet

❖ Supersaturation is one of the conditions in order to form a cloud droplet of liquid water:

❖ Supersaturation is defined as:

“The condition exists in a given portion of the atmosphere when the relative humidity is greater than 100%.”



- The predominant and more effective way to form a cloud droplet need a tiny particle or aerosol, known as a Condensation Nucleus.
- Cloud condensation nuclei (CCNs- cloud seeds) are small particles typically 0.2 μm , or 1/100th the size of a cloud droplet on which water vapor condenses.
- CCN can be made of natural materials such as **dust or volcanic ash**, or can be made of man-made pollutants such as **smoke particles**.

Source of Condensation Nuclei (CN)

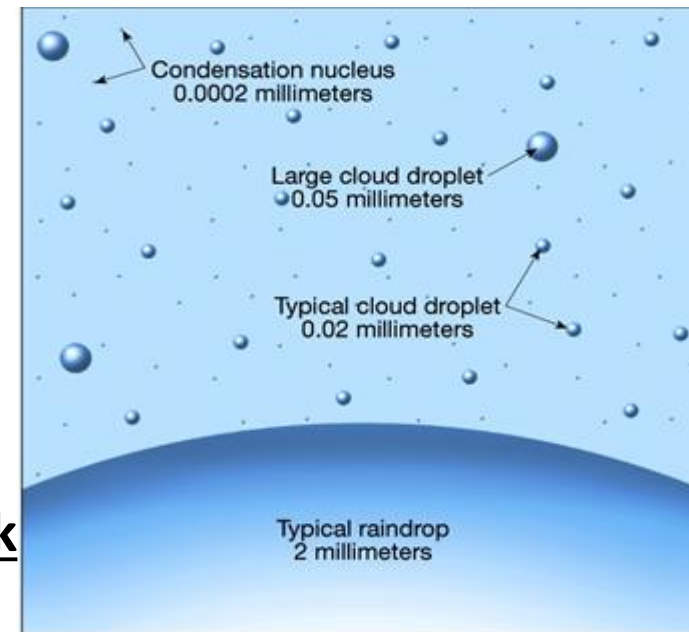
- ☐ Dust
- ☐ Volcanoes
- ☐ Smoke
- ☐ Forest Fires
- ☐ Ocean Salts/Spray
- ☐ Anthropogenic emissions

Warm Rain Formation:- Collision-Coalescence Process

- Cloud droplets initially grow by condensation, then through collision-coalescence (sticking together).
- The **collision-coalescence** process is an important mechanism in forming raindrops in **warmer clouds**

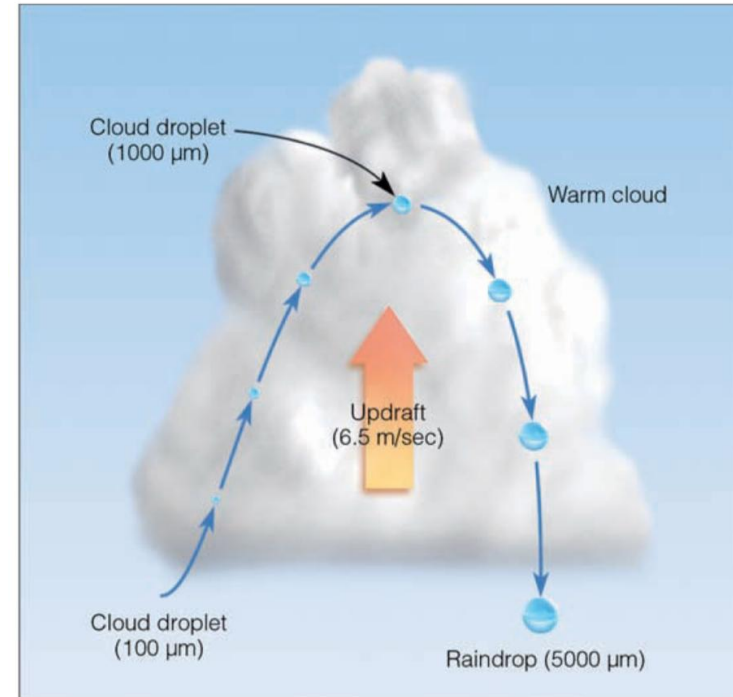
Things may happen during Collision:

- They may bounce apart.
- They may coalesce and remain permanently united.
- They may coalesce temporarily and separate, apparently retaining their initial identities.
- They may coalesce temporarily and then break into a number of small drops.



Warm Rain Formation:- Collision-Coalescence Process

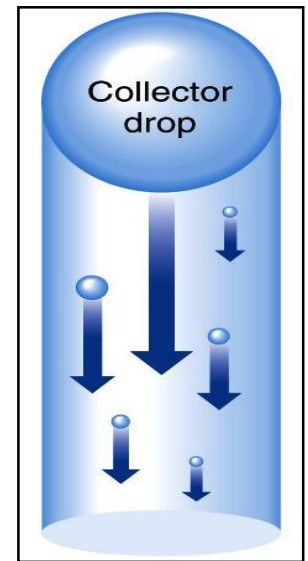
- Suppose a cloud droplet of $100\text{ }\mu\text{m}$ is caught in an updraft whose velocity is 6.5 m/sec . As the droplet rises, it collides with and captures smaller drops in its path and grows until it reaches a size of about $1000\text{ }\mu\text{m}$.
- At this point, the updraft in the cloud is just able to balance the pull of gravity on the drop.
- Here, the drop remains suspended until it grows just a little bigger. Once the fall velocity of the drop is greater than the updraft velocity in the cloud, the drop slowly descends.
- As the drop falls, larger cloud droplets are captured by the falling drop, which then grows larger. By the time this drop reaches the bottom of the cloud, it will be a large raindrop with a diameter of over 5 mm .



● **FIGURE 7.5** A cloud droplet rising then falling through a warm cumulus cloud can grow by collision and coalescence, and emerge from the cloud as a large raindrop.

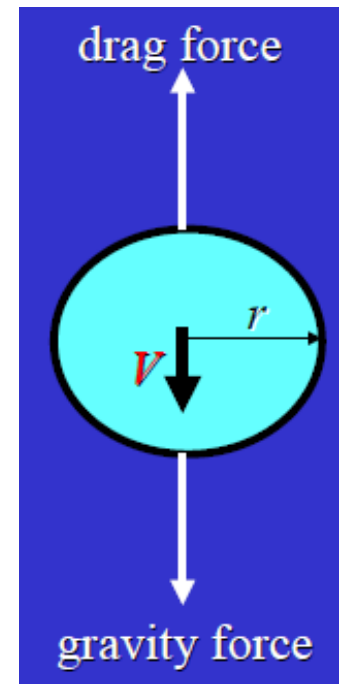
- Because raindrops of this size fall faster and reach the ground first, they typically occur at the beginning of a rain shower originating in these warm, convective cumulus clouds.

The largest droplet (collector drop) falls through a warm cloud and overtakes some of the smaller droplets because of its greater terminal velocity contributing to the collision–coalescence process.



Terminal Velocity

- Terminal velocity is the constant speed that a falling object has when the gravity force and the aero dynamical drag force applied on the subject reach a balance.
- Terminal velocity depends on the size of the object: small objects fall slowly and large objectives fall quickly.



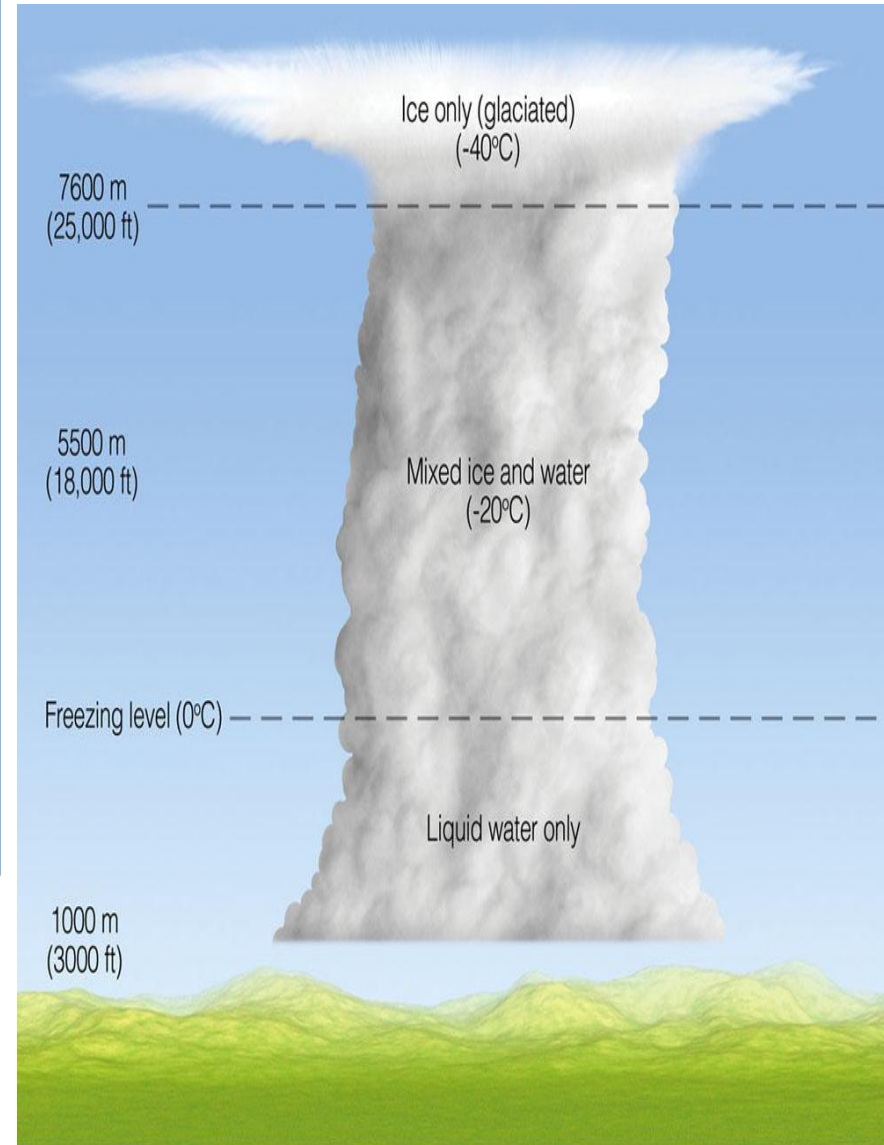
Cold Rain Formation-Bergeron-Findeisen Process

- ❖ Can be comprised of:
 - ✓ Water
 - ✓ super-cooled water & liquid droplets observed at temperatures less than 0°C
 - ✓ Ice
- ❖ Super cooled water is found at altitudes where:
 $-40^{\circ}\text{C} < \text{Temp} < 0^{\circ}\text{C}$
- ❖ At altitudes above -40°C , only ice is found
- ❖ Dominating rain making process is “Bergeron- Findeisen process”

Ice Crystal Process

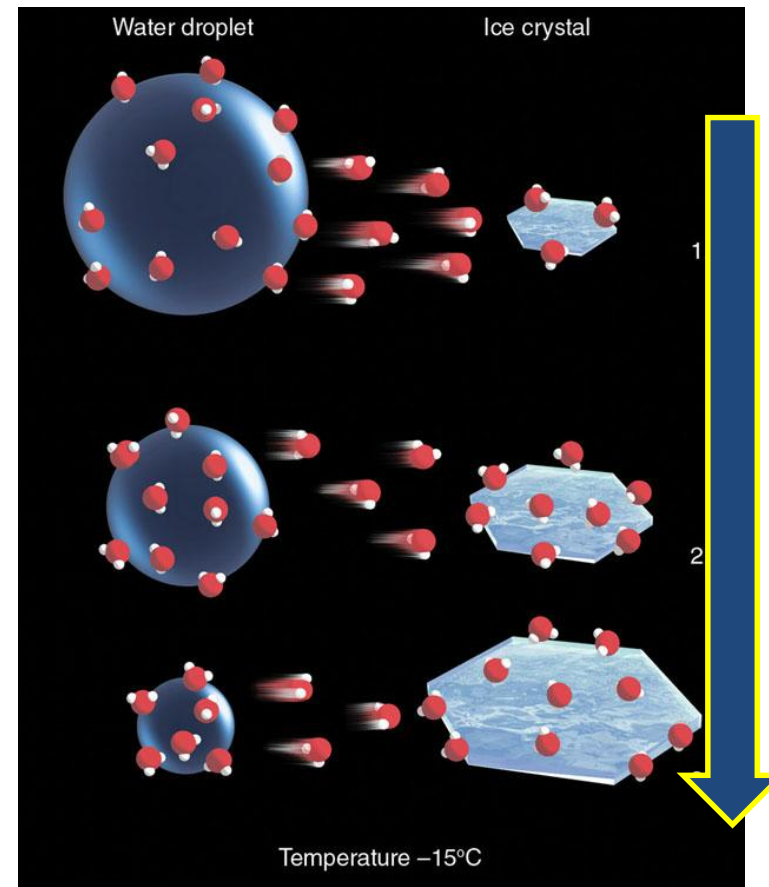
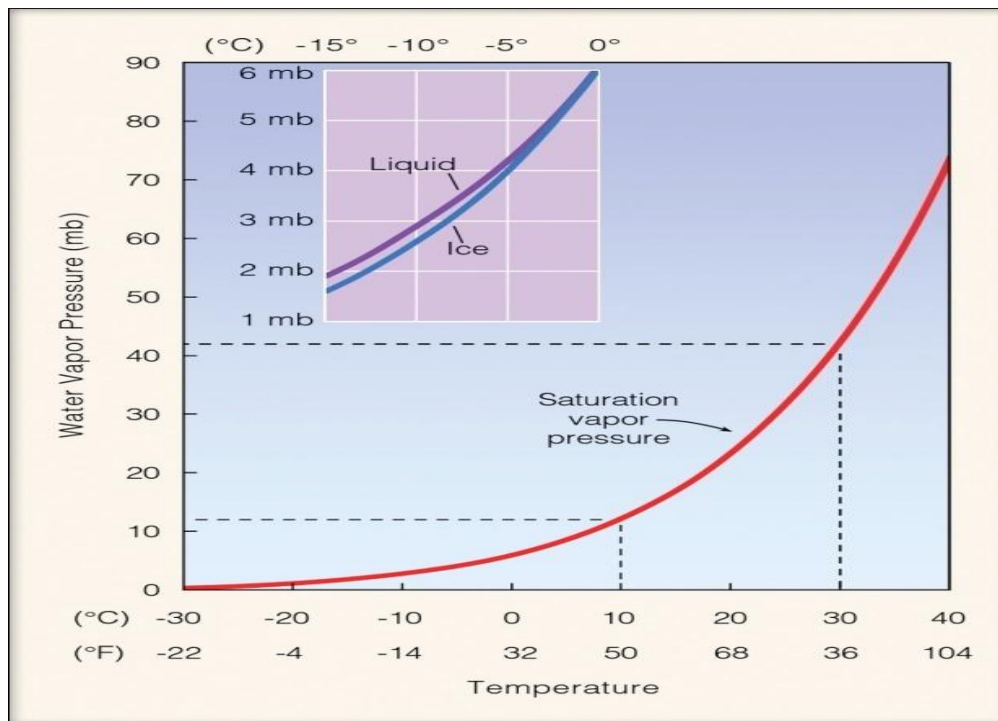
Key Concept

Saturation vapor pressure_{ice} is less than
Saturation vapor pressure_{water}



Bergeron-Findeisen Process or Ice Crystal Process

- Saturation vapor pressures over liquid water is higher than over ice.
- This causes water vapor molecules to diffuse from water droplet towards the ice crystals.
- Ice crystals grow at the expense of water droplets. It is called ice-crystal (or Bergeron) process.



In mixed-phase clouds, ice grows rapidly at the expense of the liquid because the saturation vapor pressure is lower with respect to ice than water and the process known as the Bergeron-Findeisen process.

Bergeron-Findeisen Process or Ice Crystal Process

Step 1: The difference in vapor pressure causes water vapor molecules to move(diffuse) from the droplet toward the ice crystal.

Step 2: The removal of vapor molecules reduces the vapor pressure above the droplets.

Step 3: Droplet is now out of equilibrium with its surrounding, it evaporates to replenish the diminished supply of water vapor above it.

Step 4: This process provides a continuous source of moisture for the ice crystal which absorbs the water vapor and grows rapidly (Accretion/Riming) .

Step 5: Hence ice crystals grow larger at the expense of the surrounding water droplets.

Convective and Stratiform

Precipitation produces in two types, depending upon the dominant mechanism responsible for their generation.

(e.g. Vertical motion)

Convective precipitation

- Convective clouds (e.g. Cumulonimbus)
- High rain rate,
- Strong horizontal reflectivity gradients
- Large vertical velocity;
- Coalescence

Stratiform precipitation

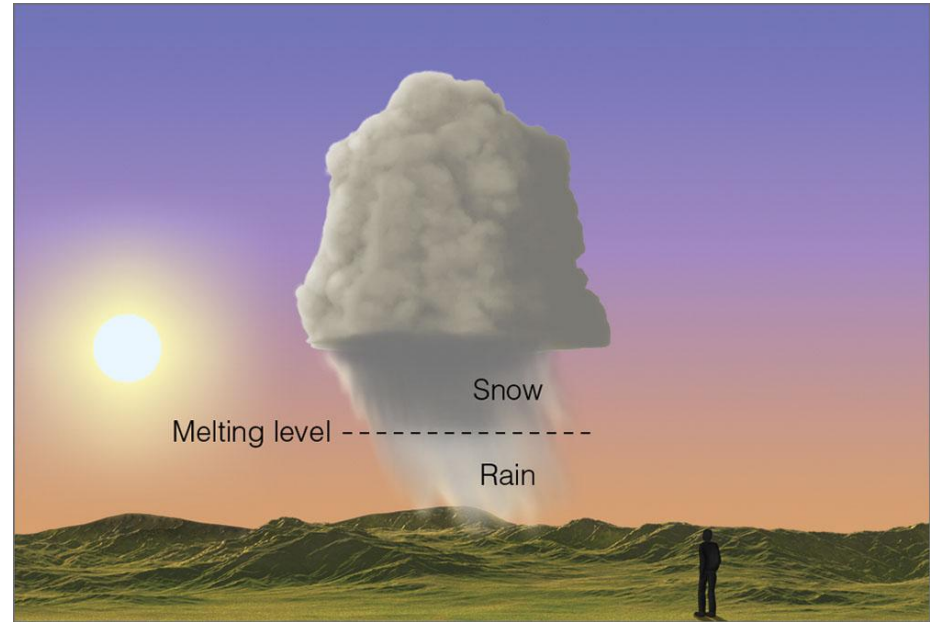
- Stratiform Clouds (e.g. Nimbostratus):
- Light to moderate rain rate, life cycle is more than convective
- Weak horizontal reflectivity gradients,
- Weak vertical velocity.
- Vapor deposition

Precipitation Types

- Rain
- Snow
- Sleet and freezing rain
- Hail

Rain: drop diameter of 0.5 ~ 6mm

- Drizzle: drop diameter < 0.5mm
- Virga: rainfall not reaching surface



1



2



3

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Rain drop shape

1: Tear-shaped

2: Spherical shape for small raindrops with diameter < 2 mm

3: For large raindrops with diameter > 2 mm- Kidney bean

Snow

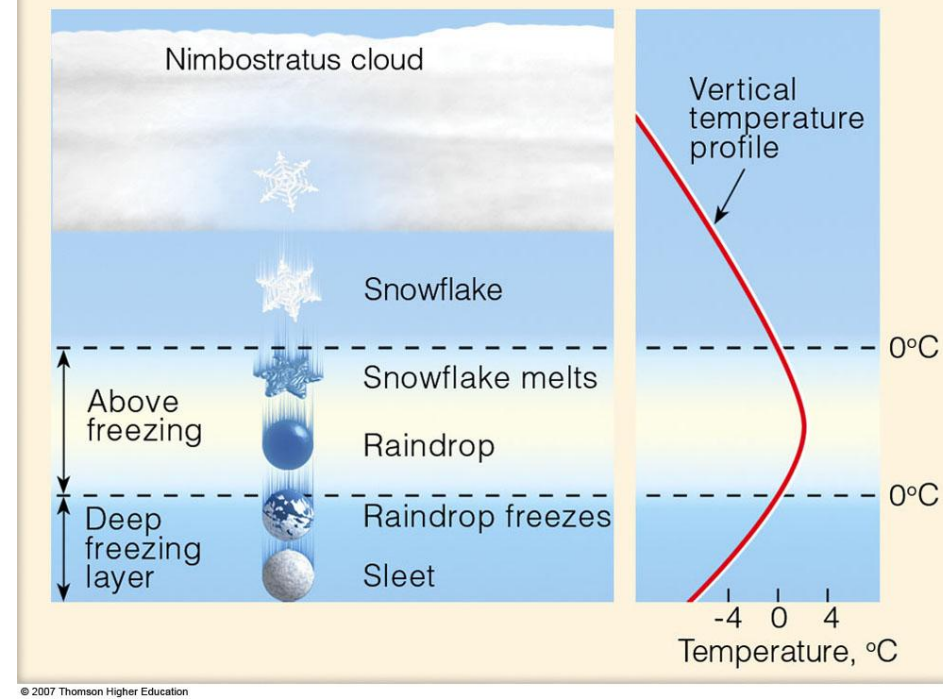
We know that much of the precipitation reaching the ground actually begins as snow.

- **Fall streaks:** ice crystals and snowflakes from high cirrus clouds that usually do not reach surface.
- **Flurries:** light snow falling from cumulus clouds
- **snow storm:** heavy snowfall
- **Blizzard:** low Temp and strong wind bearing large amounts of snow, reducing visibility to a few meters



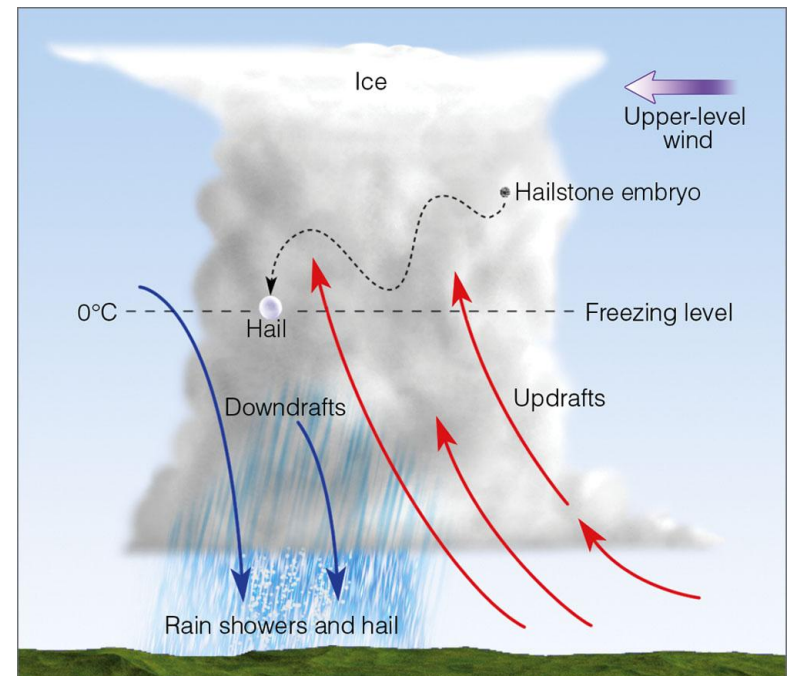
Sleet and Freezing Rain

- **Sleet**: frozen raindrop (called ice pellet); makes a tapping sound when falling on glass; bounce when striking the ground.
- **Freezing rain**: **Supercooled** liquid drops spread out and freeze on cold surface



Hail

- Hail is produced when graupel grows by accumulating supercooled liquid droplets- “a process called accretion”.
- **Strong updrafts** are needed.



Examples of Precipitation Types & drop sizes

<u>Type</u>	<u>Size</u>	<u>Description</u>
Drizzle	< 0.5 mm	Small uniform drops that fall from stratus clouds
Rain	0.5 - 5 mm	Size of drops generally vary from one place to another
Freezing Rain	0.5 - 5 mm	Rain that freezes on contact with object
Sleet	0.5 - 5 mm	Ice particles from raindrops that freeze during descent
Snow	1 - 2 mm	Aggregated ice crystals that remain frozen during descent
Hail	5 to 10 cm or larger	Hard pellets of ice from cumulonimbus clouds

Hydrometeors

- Hydrometeors: Any product of **condensation** or **deposition** of water vapor, whether formed in the free atmosphere or at the earth's surface
- It is consisting of an **ensemble of liquid or solid water particles suspended in or falling** through the atmosphere, **blown** by the wind from the Earth's surface, or **deposited** on the objects on the ground or in the free air.

Basically, the hydrometers can:

- Fall through the atmosphere.
- Be suspended in the atmosphere.
- Be blown from the ground into the atmosphere by the wind.
- Be deposited on objects.

Kinds of Hydrometeors

Hydrometeors

Graupel

Drizzle

Rain water

Snow

Cloud liquid Water

Rain Water

Cloud Ice

Snow Grains

Snow pellets

Ice Pellets

Hail

Hailstones

Ice crystals

Fog

Mist(Ground Fog)

Drifting or Blowing snow

Cloud droplets

Rain

Dew

Hoar Frost

Glaze (Clear Ice)

Precipitating Ice

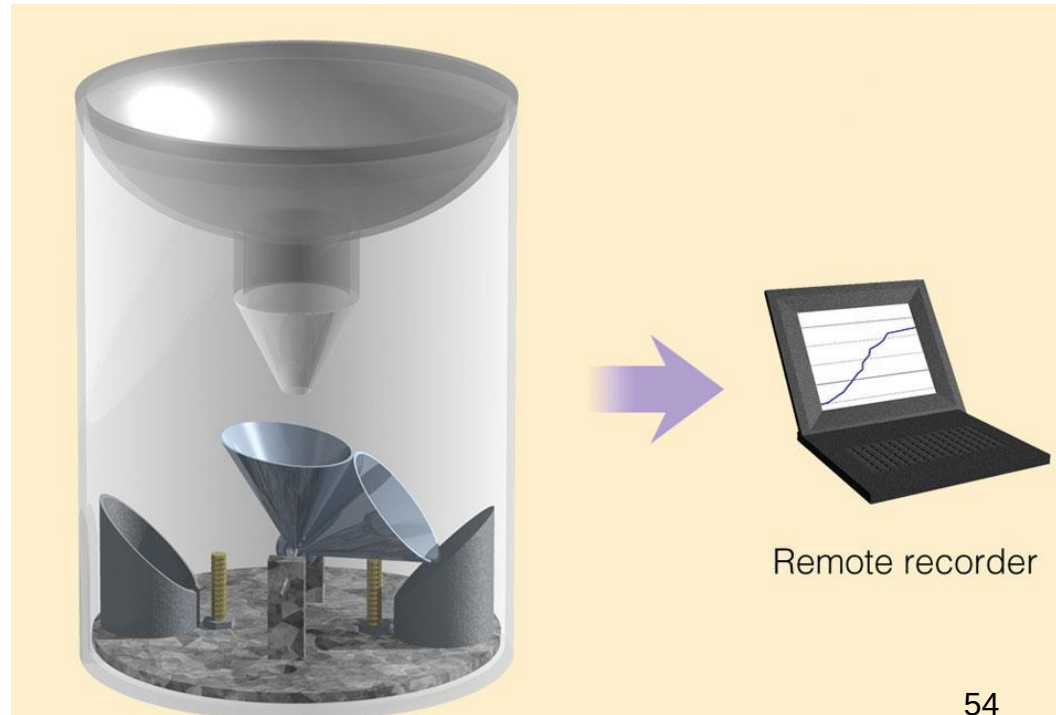
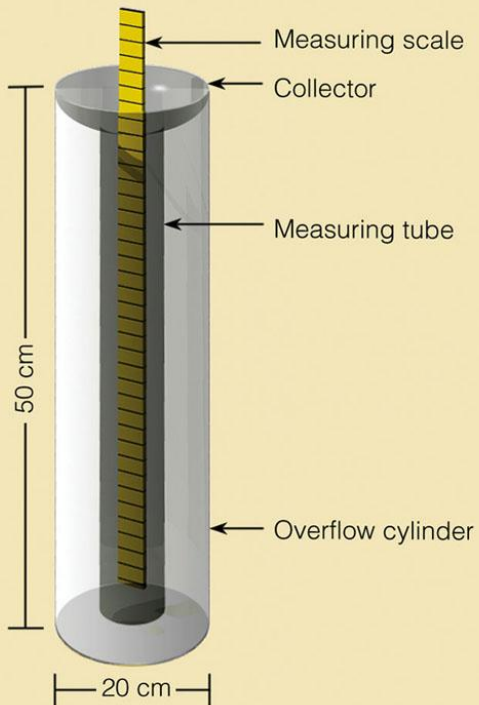
Kinds of Hydrometeors

- **Drizzle:** It is the precipitation of liquid water particles in the form of droplets with diameter of less than 0.5 mm.
- **Rain:** It is the precipitation of liquid water particles in the form of drops of more than 0.5 mm in diameter.
- **Snow:** It is the precipitation of ice crystals. Many of them are branched structures, sometimes in the shape of stars.
- **Snow Pellets:** It is the precipitation of small snow grains, white and opaque, generally round. Their diameter is smaller than 1 mm.
- **Ice Pellets:** It is the precipitation of transparent or translucent little balls of ice, with a diameter of 5 mm or less.
- **Hail:** It is the precipitation of ice balls ranging in diameter from 5 to 50 mm, sometimes even more.

Precipitation Measurement Instruments

❖ Standard rain gauge:

❖ Tipping bucket rain gauge:

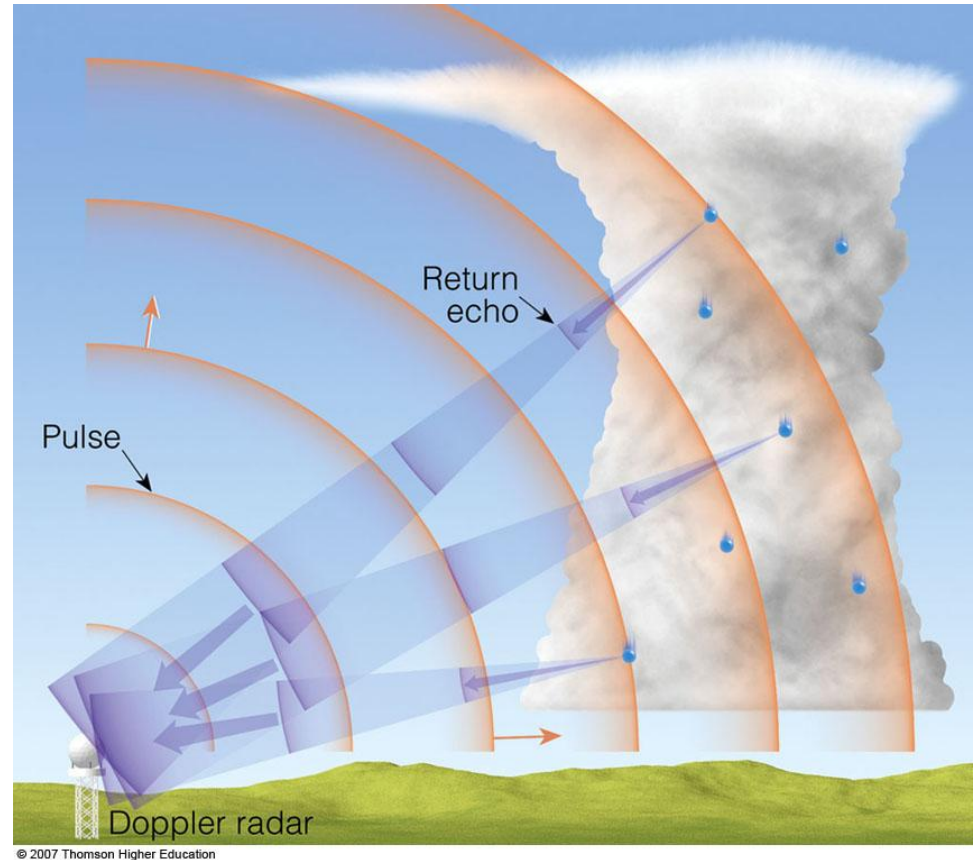


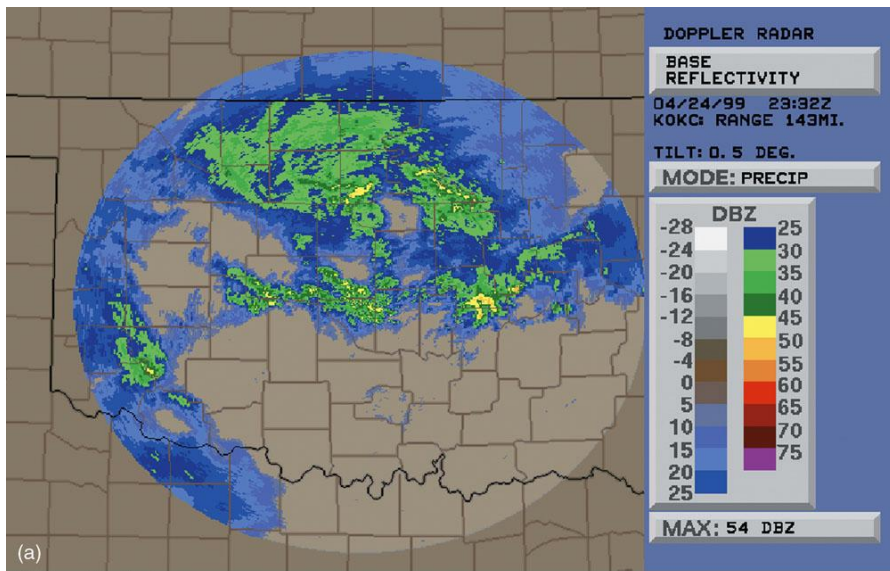
Doppler Radar and Precipitation

Radar: radio detection and ranging Doppler radar:

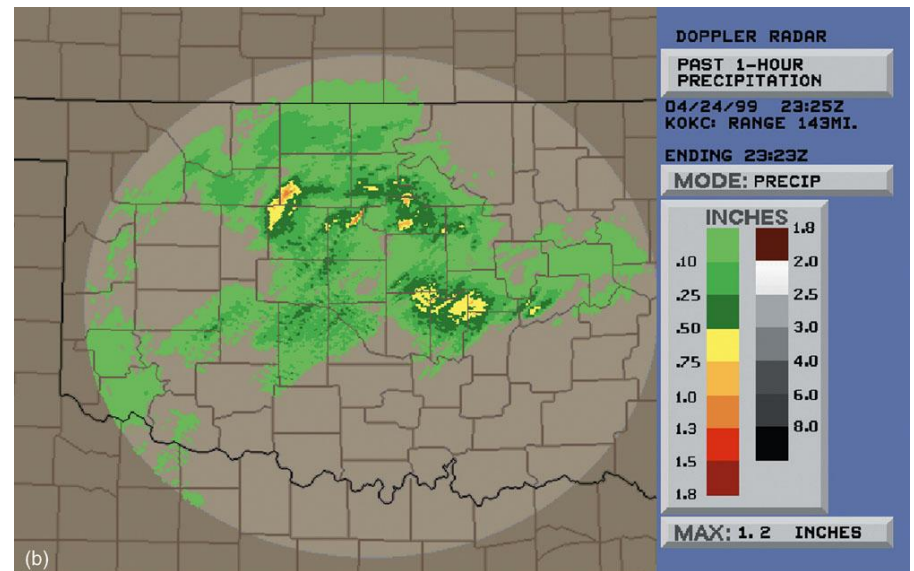
Uses Doppler shift
(e.g., a higher-pitched whistle as a train approaches you);

- Provide precipitation area and intensity;
- Provide horizontal speed of falling rain





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Measuring precipitation from space.

“Remote Sensing”

E.g.: TRMM

